

## Interview Questions

*Note:*

The codes will be evaluated on the results, cleanliness, and explanatory comments. On each section you can add any information which you think it helps audiences to gain better insight. Make justifiable assumptions where necessary.

### Task 1:

Task\_2.csv is the traffic data belong to a hosting provider for one day. Please provide the Python snippets to answer the following:

- The top 3 high traffic IPs after 6pm
- The 'highest traffic increase' for an hour span.
- Top 3 domain in terms of total traffic. Domain can be obtained through first two parts (Highlighted) of an IP (172.64.205.7).
- The traffic movement for top 5 IPs in each hour through the day.
- Modify the code to be memory and time efficient and providing the explanations.

### Task 2:

Use the traffic data provided in the attached traffic\_data.csv file, to create an airflow DAG file to implement the following tasks (each task has to be implemented separately). Airflow only runs once a day at midnight:

- Filter out 20% of ips (for example low traffic ips)
- Create a branch task for AM and PM traffic
- Find top 3 traffic ips in the PM branch and send the results to the selected email address (Outlook)
- Stop the process for the AM branch if the execution date is not weekend.
- Otherwise send the top 3 traffic ips to the same email above.

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- Term of Confidentiality: Indefinite.
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